



Assignment No 3

Title: Fruit Shop Management System

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Overview:

I developed a management system for a fruit shop. It helps the shopkeeper to maintain the records of Fruits and update and enter new records.

Implementation:

I created a structure named item containing 4 data members

- Name type char array (max size 20)
- Code type int.
- Price type float.
- Stock type int.

Then, I created a class named Fruit which contains only 1 private data member which of `std::list<item>` type and following are my functions

- `void load_data();`
- `std::list<item>::iterator find(int code=-1,const char[]="dummy");`
- `void del();`
- `void add();`
- `void update();`
- `void import(const std::string & filepath);`
- `void Export(const std::string & filepath);`
- `void append_data(int curr_size);`
- `int get_prev_size();`
- `void write_code(int code);`
- `bool code_validate();`
- `void del_code(int code);`
- `void search();`
- `void display();`
- `void showHelp();`

void load_data()

in this function I read all the data from the binary files to data member fruit.

int get_prev_size()

This function return the number of items stored in the binary file.

Void write_code(int code)

This function takes a code as a parameter and appends this code into the binary file.

Void del_code(int code)

This function loads all the codes from the binary file and find the code and deletes it from the array and rewrites the array into the binary file.

Bool code_validate(int code)

This function checks that if the code in the parameter matches any of the previous codes or not.

Void add()

This function helps the user to add new items.

Void del()

This function help the user to delete a specific record.

Std::list<item>::iterator find(int code=-1,const char name="dummy")

This function takes the code or name of the item and return the iterator pointing on it.

Void search()

This function allows user to find a specific record.

Void update()

This function allow user to update a specific attribute in a specific record.

Void del()

This function allows user to remove/delete a specific record from the list and binary file.

Void import(const std::string& filepath)

This function reads the data from the file of type csv and writes it into the list.

Void export(const std::string& filepath)

This function writes the data from list to the specified csv file given in the parameter.

Void showHelp()

This function guides the user how to interact with the system through command line. It displays the instructions from a Text File.

How to Use?

System can be operated through command line. Different commands for different options are given below

- 1- Display (info.exe display)**
- 2- Delete (info.exe delete)**
- 3- Update (info.exe update)**
- 4- Add (info.exe add)**
- 5- Import (info.exe import filename.csv)**
- 6- Export (info.exe export filename.csv)**
- 7- Search (info.exe search)**

To get help user can use command (**info.exe help**)

If no argument/command will be given then a main menu will be opened as following:

***** Welcome to Saddam Fruit Market *****

Search Item [S/s]

Add Item [A/d]

Delete Item [D/d]

Update Item [U/u]

Export Items [E/e]

Import Items [I/i]

Display Items [P/p]

Show Help [H/h]

Terminate [X/x]

User can input the option for his/her desired operation.